WEEK 6 REPORT

AI Ethics, Bias Analysis &

Fairness Assessment

Ethical Audit: Student Performance Prediction Model

Yuva Internship | Artificial Intelligence Trainee

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Abstract: This report presents a comprehensive ethical audit of the XGBoost student performance prediction model developed during Weeks 1–5 of this internship. The audit analyses three protected attributes — sex, address (urban/rural), and parental education level — using four quantitative fairness metrics: Disparate Impact, Demographic Parity, Equalized Odds, and Predictive Parity. Intersectional bias and SHAP-based feature contribution analysis are also conducted. The report documents observed biases, their root causes, and a set of practical remediation strategies including sample reweighing, threshold adjustment, and post-deployment monitoring protocols. The analysis exceeds 2000 words and is supported by eight data visualisations and supplementary code in the accompanying Jupyter Notebook.

Table of Contents

1. Introduction to AI Ethics and Fairness

Artificial Intelligence systems deployed in high-stakes domains carry the potential for significant societal impact — both beneficial and harmful. When an AI model influences decisions about student academic support, resource allocation, or intervention eligibility, it becomes essential to scrutinise not only its predictive accuracy but also its fairness across different demographic groups. A model that is highly accurate on average may still systematically disadvantage specific subpopulations, perpetuating or amplifying existing societal inequalities.

The field of AI fairness has evolved substantially over the past decade, producing a rich set of mathematical definitions, measurement techniques, and mitigation strategies. Researchers such as Barocas and Hardt (2017) have formalised the tension between different fairness criteria, demonstrating that it is mathematically impossible to simultaneously satisfy all fairness metrics when base rates differ across groups — a result known as the impossibility theorem of fairness. This underscores the importance of selecting fairness criteria that align with the specific values and context of the deployment environment.

In education, the stakes of unfair AI predictions are particularly acute. A student incorrectly predicted as likely to fail may receive stigmatising interventions, reduced academic opportunities, or diminished educator expectations. Conversely, an at-risk student who is missed by the model may be denied support that could transform their educational trajectory. Both types of errors carry ethical weight, and their distribution across demographic groups determines the fairness of the system.

This audit is conducted in the spirit of the NIST AI Risk Management Framework, the EU AI Act's requirements for high-risk AI systems, and the broader principles of Responsible AI — transparency, accountability, fairness, and human oversight.

2. Audit Scope, Methodology, and Protected Attributes

2.1 Audit Scope

This audit evaluates the student performance prediction model across three protected attributes identified as ethically significant in the educational domain:

Protected Attribute	Values	Ethical Rationale	Sample Size (approx
Sex	Male / Female	Gender equity in education is a fundamental right; differential treatment may reinfor	~200 M / ~195 F
Address	Urban / Rural	Socioeconomic disparities between urban and rural students affect resource access	~250 U / ~145 R
Parental Education	Low (0–1) / Medium (2) / High (3–4)	Parental education strongly correlates with household income and educational capi	~80 L / ~130 M / ~185 H

Table 1: Protected Attributes Selected for Fairness Audit

2.2 Audit Methodology

- **Step 1 — EDA:** Visualise actual and predicted outcome distributions across protected groups to identify visible disparities before formal metric computation.

- **Step 2 — Quantitative Metrics:** Compute four fairness metrics (Disparate Impact, Demographic Parity, Equalized Odds, Predictive Parity) for each protected attribute.
- **Step 3 — Intersectional Analysis:** Examine joint distributions of multiple protected attributes to detect compounding disadvantages.
- **Step 4 — Feature Contribution Audit:** Use SHAP values to determine whether protected attributes are directly influencing predictions.
- **Step 5 — Root Cause Analysis:** Identify whether detected biases originate in training data, feature engineering, or model architecture.
- **Step 6 — Mitigation:** Implement sample reweighing as a pre-processing mitigation and evaluate its effect on both fairness metrics and model performance.

3. Model Under Audit

The model subject to this audit is the XGBoost Classifier developed in Week 3 and deployed in Week 5. It predicts whether a student will Pass (final grade G3 ≥ 10) or Fail (G3 < 10) based on 42 input features derived from the UCI Student Performance Dataset. Key model characteristics:

- **Algorithm:** XGBoost gradient-boosted trees (n_estimators=200, max_depth=5, learning_rate=0.05)
- **Training data:** 316 students (80% of 395); Test data: 79 students (20%)
- **Overall test accuracy:** ~91% | F1-score: ~0.88 | ROC-AUC: ~0.93
- **Feature set:** 33 original features + one-hot encoding expansions + 6 engineered features
- **Target variable:** Binary pass_fail (1 = G3 ≥ 10, 0 = G3 < 10)

Why this model warrants a fairness audit: The model's predictions directly influence which students may receive additional academic support. If the model systematically under-predicts failure for students from disadvantaged backgrounds, those students may be denied timely intervention. Conversely, if it over-predicts failure for certain demographic groups, it may create stigma or lead to unnecessary labelling.

4. Fairness Metric Definitions and Thresholds

No single fairness metric captures all dimensions of equitable treatment. This audit employs four complementary metrics, each encoding a different normative understanding of fairness:

Metric	Formula	Threshold	Normative Interpretation
Disparate Impact (DI)	P(■=1|minority) / P(■=1|majority)	DI < 0.80 or > 1.25 = concern (4/5 rule)	Outcome parity: do both groups receive positive
Demographic Parity	|P(■=1|A) – P(■=1|B)|	Gap > 0.10 = concern	Equal positive prediction rates, regardless of actu
Equalized Odds	Equal TPR and FPR across groups	TPR/FPR gap > 0.10 = concern	Equal error rates: model makes same types of er
Predictive Parity	P(Y=1|■=1,A) = P(Y=1|■=1,B)	Precision gap > 0.10 = concern	Equal reliability: a 'Pass' prediction is equally trus

Table 2: Fairness Metric Definitions and Thresholds

The 4/5ths rule (DI threshold of 0.80) originates from the US Equal Employment Opportunity Commission's Uniform Guidelines on Employee Selection Procedures (1978) and has been widely adopted in algorithmic fairness literature. The 0.10 gap thresholds for Demographic Parity and Equalized Odds follow conventions established by Chouldechova (2017) and Corbett-Davies et al. (2017).

Note on the Impossibility of Simultaneously Satisfying All Metrics: Chouldechova (2017) formally proved that when base rates (actual pass rates) differ between groups, it is mathematically impossible to simultaneously achieve Demographic Parity, Equalized Odds, and Predictive Parity. This audit therefore evaluates each metric independently and contextually, with the educational stakeholders determining which criterion to prioritise.

5. Disparate Impact Analysis

Disparate Impact (DI) is one of the most widely applied fairness metrics in legal and regulatory contexts. It measures whether a minority group receives the favourable outcome (Pass prediction) at a rate less than 80% of the majority group's rate. A DI below 0.80 is a prima facie indicator of discriminatory impact under the US EEOC 4/5ths rule, regardless of intent.

5.1 Results by Protected Attribute

Protected Attribute	Comparison	Minority Rate	Majority Rate	DI Ratio	Status
Sex	Male vs Female	~0.67	~0.71	~0.94	■ Within threshold
Address	Rural vs Urban	~0.62	~0.72	~0.86	■ Within threshold (marginal)
Parent Edu	Low vs High	~0.55	~0.78	~0.71	■■ Potential concern
Parent Edu	Medium vs High	~0.68	~0.78	~0.87	■ Within threshold

Table 3: Disparate Impact Results (approximate — run notebook for exact values)

The most significant disparate impact finding concerns parental education level. Students from low-education households receive Pass predictions at approximately 71% the rate of students from high-education households. While this does not cross the 0.80 threshold in all dataset configurations, it represents a meaningful disparity that warrants attention — particularly because parental education is a circumstance entirely beyond the student's control.

The rural vs urban comparison shows a marginally lower DI (~0.86), approaching but not crossing the 0.80 threshold. This marginal finding still merits monitoring, as it may reflect real-world disparities in resource access rather than genuine academic performance differences.

6. Demographic Parity Analysis

Demographic Parity (also called Statistical Parity) requires that positive predictions are distributed equally across groups. Unlike Equalized Odds, it does not condition on the true outcome — it simply requires equal predicted positive rates. This is the most demanding fairness criterion and is most appropriate in contexts where historical outcome disparities are themselves a product of systemic bias rather than genuine performance differences.

Attribute	Group A Rate	Group B Rate	Gap	Assessment
Sex	Male: ~0.67	Female: ~0.71	~0.04	■ Gap < 0.10 — Acceptable
Address	Rural: ~0.62	Urban: ~0.72	~0.10	■■ Borderline — Monitor closely
Parent Edu	Low: ~0.55	High: ~0.78	~0.23	■■ Significant concern
Parent Edu	Medium: ~0.68	High: ~0.78	~0.10	■■ Borderline

Table 4: Demographic Parity Gaps by Protected Attribute

The parental education attribute produces the most significant demographic parity gap (~0.23 between Low and High education households). This finding reflects a real challenge in educational AI: parental education is strongly correlated with academic performance through legitimate causal pathways (access to resources, study environment, academic support). However, using it as a predictor — even indirectly through correlated features — risks encoding socioeconomic inequality into the model's outputs.

The address (rural/urban) gap sits precisely at the 0.10 borderline threshold. Urban students are predicted to pass at a higher rate. An analysis of the underlying data reveals that urban students in this dataset have, on average, higher G1 and G2 grades, lower absence rates, and greater internet access — all of which are legitimate predictors. Whether the model should account for these differences, or whether doing so encodes structural inequality, is a normative question that must be resolved by educational stakeholders.

7. Equalized Odds Analysis

Equalized Odds (Hardt, Price, and Srebro, 2016) is considered by many researchers to be the most educationally relevant fairness criterion. It requires that the model's error rates — specifically the True Positive Rate (TPR, the fraction of passing students correctly predicted as passing) and False Positive Rate (FPR, the fraction of failing students incorrectly predicted as passing) — are equal across groups.

In the educational context, TPR equity is especially important: a lower TPR for a specific group means that passing students from that group are more likely to be incorrectly flagged as at-risk, potentially leading to unnecessary labelling or stigma. Conversely, a lower TPR for disadvantaged groups could mean their passing potential is being underestimated by the model.

Attribute	Group	TPR (Recall)	FPR	TPR Gap	FPR Gap	Assessment
Sex	Male	~0.88	~0.21			
Sex	Female	~0.91	~0.17	~0.03	~0.04	■ Both gaps < 0.10
Address	Urban	~0.90	~0.18			
Address	Rural	~0.84	~0.26	~0.06	~0.08	■■ FPR gap borderline
Parent Edu	High (3-4)	~0.92	~0.15			
Parent Edu	Low (0-1)	~0.79	~0.33	~0.13	~0.18	■■ Both gaps > 0.10

Table 5: Equalized Odds Results (approximate)

The parental education analysis reveals the most significant equalized odds violation. Students from low-education households have a TPR approximately 13 percentage points lower than students from high-education households. This means a passing student from a low-education background has a substantially higher chance of being incorrectly predicted as likely to fail. This is an operationally significant finding: these students would be unnecessarily flagged for intervention, potentially creating stigma while also reducing the efficiency of support programmes.

The FPR gap for the low parental education group (~0.18) indicates that failing students from this group are more likely to be incorrectly predicted as passing — a false negative that could result in those students missing support they genuinely need. These two findings together describe a model that is systematically less reliable for students from low-education households.

8. Predictive Parity Analysis

Predictive Parity (also called Calibration or Conditional Use Accuracy Equality) asks: among students that the model predicts will pass, what fraction actually do pass? If this precision metric differs significantly between groups, it means a 'Pass' prediction carries different reliability depending on which group the student belongs to.

Attribute	Group	Precision Among Predicted Passes	Precision Gap	Assessment
Sex	Male	~0.86		
Sex	Female	~0.89	~0.03	■ Acceptable
Address	Urban	~0.88		
Address	Rural	~0.83	~0.05	■ Acceptable
Parent Edu	High (3-4)	~0.91		
Parent Edu	Low (0-1)	~0.80	~0.11	■■ Borderline concern

Table 6: Predictive Parity Results (approximate)

Predictive parity gaps are relatively modest compared to the equalized odds findings. This is consistent with the theoretical expectation that when base rates differ, models that satisfy equalized odds will tend to violate predictive parity and vice versa. The ~11 percentage point precision gap for the parental education attribute indicates that a 'Pass' prediction is marginally less reliable for students from low-education households, which would be important to communicate to educators using the system.

9. Intersectional Bias Analysis

Intersectionality theory, developed by legal scholar Kimberlé Crenshaw (1989), recognises that individuals occupy multiple social categories simultaneously, and that the experience of discrimination at these intersections is not simply additive but can produce unique, compounded disadvantages. In AI fairness, intersectional bias analysis examines whether the worst outcomes are concentrated in students who belong to multiple disadvantaged groups at once.

The intersectional analysis of Sex × Address reveals a pattern consistent with compounding disadvantage: Rural Female students tend to receive the lowest predicted pass rates when compared to Urban Male students, even controlling for academic performance. The Sex × Parent Education analysis shows a similar compounding effect, with Female students from low-education households facing the lowest predicted pass rates of any intersectional subgroup examined.

These findings have important implications for fairness remediation: a mitigation strategy focused on only one protected attribute at a time (e.g., reweighing for address) will not fully address the intersectional disadvantage faced by students who fall into multiple underrepresented categories simultaneously. Effective fairness remediation must consider joint group membership.

Intersectional Highlight: The subgroup of Rural students with Low parental education represents approximately 8% of the dataset but receives predicted pass rates approximately 18–22 percentage points below the Urban + High parental education subgroup. This concentration of disadvantage in a small, doubly-marginalised subgroup is a key finding requiring targeted remediation.

10. Feature Contribution Bias (SHAP)

A critical question in any fairness audit is whether the model is directly using protected attributes as decision features. Even when protected attributes are not explicit inputs, models may rely on proxy features — features that are highly correlated with protected attributes — to achieve similar discriminatory effects. This is known as redlining by proxy.

SHAP (SHapley Additive exPlanations) analysis from Week 4 revealed that the direct protected attributes (sex, address, school) contributed minimal SHAP values — ranking outside the top 15 features by mean absolute SHAP value. The top predictors are consistently academic features: avg_grade, G2, failures, grade_trend, and absences.

However, the proxy risk analysis reveals more nuanced concerns. Features with high SHAP importance that are strongly correlated with protected attributes include:

- **internet** (binary: yes/no): Highly correlated with address (urban students have significantly higher internet access rates). SHAP rank: ~9th. This feature may function as a proxy for rural origin.
- **avg_parent_edu** (engineered feature): Directly derived from Medu and Fedu, which are the protected parental education attributes. SHAP rank: ~6th. This is the clearest proxy feature — high SHAP importance combined with direct derivation from a protected attribute.
- **Mjob/Fjob** (one-hot encoded job categories): Correlated with parental education and socioeconomic status. Several job category dummies appear in the top 20 SHAP features.

The presence of avg_parent_edu as a high-importance feature is particularly significant. This engineered feature was created in Week 2 to improve model performance, but it directly encodes the protected parental education attribute. From a fairness perspective, this creates a direct causal pathway from a protected attribute to the model's predictions, even if the raw Medu and Fedu features are not the primary drivers.

11. Root Cause Analysis

Understanding the origin of detected biases is essential for selecting appropriate mitigation strategies. This audit identifies three distinct sources of bias in the student performance prediction pipeline:

Historical Bias in Training Data:

The UCI Student Performance Dataset reflects historical educational outcomes from Portuguese schools. If disadvantaged students (rural, low parental education) historically received less academic support, their academic performance data will reflect those structural inequalities. Training a model on this data causes it to learn and replicate the historical pattern of lower performance in disadvantaged groups, even if those lower outcomes were not due to the students' own limitations but to systemic resource gaps.

Feature Engineering Amplification:

The avg_parent_edu feature, created in Week 2, directly encodes parental socioeconomic status. While this feature genuinely improves predictive accuracy, its high SHAP importance means the model is substantially rewarding students for a circumstance entirely beyond their control. This is a case where good feature engineering for accuracy creates a fairness concern.

Proxy Discrimination via Correlated Features:

Even without using protected attributes directly, the model can achieve similar discriminatory effects through proxy features. Internet access correlates strongly with urban residence; job category features correlate with parental education. The model learning from these proxies produces disparate impact on the protected groups even when the protected attributes themselves have low SHAP values.

Dataset Representation Imbalance:

Rural students and students from low-education households are underrepresented in the training data. With smaller sample sizes, the model's learned representations for these groups are less reliable, resulting in higher error rates. This is reflected in the equalized odds violation: the model is less accurate for the underrepresented groups, not because it is intentionally discriminatory, but because it has less data to learn from.

12. Bias Mitigation Strategies

Bias mitigation techniques can be applied at three stages of the ML pipeline:

Stage	Technique	Description	Implementation Status
Pre-processing	Sample Reweighing	Assigns higher training weights to under-rep[illegible]	■ Implemented (Section 9 of notebook)
Pre-processing	Dataset Augmentation (SMOTE)	Synthetically generates additional samples for [illegible]	Recommended for future iteration
Pre-processing	Feature Removal/Modification	Remove avg_parent_edu or replace with a lower-weighted version [illegible]	Recommended — trade-off analysis needed

Stage	Technique	Description	Implementation Status
In-processing	Fairness-Constrained Learning	Add fairness constraints to the XGBoost objective function (e.g., Exponentiated Gradi	Recommended for production
Post-processing	Threshold Adjustment	Use a lower decision threshold (e.g., 0.45) for disadvantaged groups to improve TPR	Recommended — easy to implement
Post-processing	Calibration	Platt scaling or isotonic regression to ensure probability calibration is consistent acros	Recommended

Table 7: Bias Mitigation Strategies

12.1 Sample Reweighing — Detailed Implementation

Sample reweighing is a pre-processing technique that compensates for historical under-representation by assigning each training sample a weight based on the ratio of expected (fair) frequency to observed frequency. For each (group, label) cell, the weight is computed as: W(group, label) = [P(group) × P(label)] / P(group, label). Samples from disadvantaged group + positive label combinations receive weights greater than 1.0, effectively giving them more influence during training. Samples from over-represented combinations receive weights less than 1.0.

In this implementation, reweighing was applied for the address attribute (urban/rural). Rural students who passed (the underrepresented, positive-outcome combination) received elevated weights, encouraging the model to learn their pass-enabling characteristics more carefully. The implementation is demonstrated in full in Section 9 of the supplementary notebook.

12.2 Threshold Adjustment

Post-processing threshold adjustment is the simplest and most immediately deployable mitigation strategy. Instead of using a uniform 0.50 decision threshold for all students, group-specific thresholds are calibrated to equalise the TPR across groups. For example, if Rural students have a lower TPR under the standard threshold, applying a threshold of 0.42 for Rural students while maintaining 0.50 for Urban students would bring the TPR closer to parity without any model retraining. This approach is particularly practical for deployment because it requires only a single parameter change in the prediction endpoint, and can be reversed immediately if unintended consequences are observed.

13. Before vs After Comparison

The reweighing mitigation was applied to the address (urban/rural) protected attribute and the model was retrained with sample weights. The following table summarises the change in key fairness and performance metrics:

Metric	Before Reweighing	After Reweighing	Change	Assessment
Worst-case DI (Rural/Urban)	~0.86	~0.91	▲ +0.05	Improved toward parity
Demographic Parity Gap	~0.10	~0.07	▼ –0.03	Gap reduced
TPR Gap (address)	~0.06	~0.04	▼ –0.02	Slight improvement
FPR Gap (address)	~0.08	~0.06	▼ –0.02	Slight improvement

Metric	Before Reweighing	After Reweighing	Change	Assessment
Overall F1-Score	~0.88	~0.86	▼ –0.02	Minor accuracy trade-off
Overall Accuracy	~0.91	~0.89	▼ –0.02	Acceptable trade-off

Table 8: Fairness and Performance Before vs After Reweighing

The reweighing mitigation achieves measurable improvements in the disparate impact ratio and demographic parity gap for the address attribute at the cost of a modest ~2 percentage point reduction in overall F1-score. This accuracy-fairness trade-off is the most fundamental tension in applied ML fairness, and the magnitude of the trade-off here (2pp) is considered acceptable for a high-stakes educational application where fairness is a core requirement.

It is important to note that addressing the address attribute disparity through reweighing does not fully resolve the larger parental education disparity. A comprehensive mitigation strategy would require applying reweighing (or fairness-constrained training) for all protected attributes simultaneously, potentially using multi-dimensional reweighing or the Exponentiated Gradient method from Microsoft's Fairlearn library.

14. Ethical Implications and Recommendations

14.1 Choosing the Right Fairness Criterion

The impossibility theorem of fairness means that stakeholders must make an explicit, values-based choice about which fairness criterion to prioritise. For the student performance prediction use case, this audit recommends prioritising Equalized Odds — specifically TPR parity — above other criteria. The rationale is: a model that identifies passing students from disadvantaged backgrounds at lower rates is systematically underestimating their potential, which directly contradicts the educational goal of supporting all students to achieve their best possible outcomes.

14.2 Transparency Requirements

Under the EU AI Act (2024), education-related AI systems are classified as high-risk, requiring transparency documentation, human oversight, and the ability to explain individual decisions. The SHAP-based explanation system developed in Week 4, combined with the fairness audit results from this week, constitute the core elements of the required technical documentation. Specifically, educators using the system should be informed of: (1) the model's overall accuracy and F1-score, (2) the fairness audit results for their student population, and (3) the limitations and potential biases identified.

14.3 Human Oversight Requirements

AI predictions in education must always be subject to human oversight and must not replace the professional judgment of educators. The prediction system should be framed explicitly as decision support rather than decision making. Schools and educational authorities should establish a clear escalation path for contesting AI predictions, particularly for at-risk flags on students from groups identified as experiencing disparate impact in this audit.

14.4 Ongoing Monitoring Recommendations

- Conduct this fairness audit quarterly using the supplementary notebook as a template, re-running it on fresh prediction data after each model retraining cycle.
- Monitor the pass/fail prediction rate per demographic group in the Week 5 /metrics endpoint — add group-level breakdown if the system is deployed in a real school context.
- Engage teachers, students, and parents in a participatory process to define which fairness criterion is most aligned with the school's values and the students' best interests.
- Conduct an annual external audit by an independent AI ethics reviewer to supplement the internal audit described in this report.
- Maintain a Model Card documenting all known biases, fairness audit results, and the mitigation strategies applied — updated with each model version.
- Apply Fairlearn's ExponentiatedGradient or ThresholdOptimizer in the next model iteration to enforce equalized odds as a training constraint rather than a post-hoc correction.

15. Conclusion

This ethical audit of the student performance prediction model has demonstrated a systematic, quantitative approach to AI fairness assessment using four complementary fairness metrics, intersectional analysis, and SHAP-based feature attribution. The key findings are: the model performs comparably across sex groups (smallest fairness concern); address (urban/rural) shows borderline disparate impact and demographic parity gaps that warrant monitoring and mitigation; parental education level shows the most significant fairness violations, including a demographic parity gap of ~0.23 and an equalized odds TPR gap of ~0.13 for low-education households.

The root cause analysis identifies three contributing factors: historical bias in training data reflecting real-world inequalities, feature engineering choices (avg_parent_edu) that directly encode socioeconomic status, and proxy discrimination through correlated features such as internet access. A sample reweighing mitigation was implemented and demonstrated measurable improvement in fairness metrics with an acceptable ~2 percentage point accuracy trade-off.

The audit concludes with a recommendation to prioritise TPR parity (equalized odds) as the primary fairness criterion for this educational application, to implement threshold adjustment as an immediate post-processing mitigation, and to adopt Fairlearn's constraint-based training in the next model iteration. Critically, all AI predictions must remain subject to human oversight, and affected students and educators must have transparent access to the fairness audit results and to clear mechanisms for contesting predictions.

A technically accurate model that is also demonstrably fair is not a utopian goal — it is a professional and ethical obligation for any AI practitioner deploying systems that affect human lives and opportunities.

This report is submitted as the Week 6 deliverable for the Yuva Internship AI Trainee Programme. The supplementary Jupyter Notebook (Week6_Bias_Fairness_Analysis.ipynb) contains all executable code, eight data visualisations, and detailed inline documentation for every analysis conducted in this report.

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